

BioNeuron

A Biologically Accurate Spiking Neural Network:
Design, Mechanisms, Simulations, and Results

Author: Jugal Kishore Nagulapalli | **Software:** bioneuron v1.8 (MIT) | **Stack:** Python 3.10, PyTorch 2.6 (CUDA)

Summary. BioNeuron is a from-scratch cortical spiking neural network in which every neuron follows biophysical rules. Learning is driven by local spike-timing plasticity and reward/stress neuromodulation, not by backpropagation. Connectivity is not fixed: synapses are grown from correlated activity, strengthened or weakened by plasticity, and pruned. The system reproduces a set of standard results from computational neuroscience (spike-timing learning, spike-frequency adaptation, short-term plasticity, gamma oscillations, self-organized criticality, dendritic nonlinearity) and learns simple tasks from its own self-generated feedback. This report states each mechanism, the equation it implements, the reference it follows, and the measured result. All numbers are from runs on the machine described in Section 9.

Design rule. Biological accuracy is the priority. A correctness fix (for example, the spike-timing sign correction in Section 5.1) is applied unconditionally. Any mechanism that changes the dynamics of existing experiments is off by default and enabled by one flag, so earlier results stay reproducible.

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1. Overview

A conventional artificial neural network is a set of continuous-valued units connected by a fixed weight matrix trained by gradient descent. BioNeuron is different on all three counts. Units are spiking neurons with a membrane voltage that integrates input and emits discrete spikes. Weights change by local rules that depend only on the relative timing of pre- and post-synaptic spikes and on broadcast chemical signals. The wiring itself is dynamic: it starts sparse and random, and is reshaped by activity.

The model is organized as composable layers, each grounded in a specific piece of biology:

- **Membrane dynamics** - leaky integrate-and-fire, optionally extended to the adaptive exponential (AdEx) model, with spike-frequency adaptation.
- **Receptors and channels** - fast (AMPA-like) and slow voltage-gated NMDA excitation, GABA inhibition, gap junctions, and a two-compartment active dendrite.
- **Cell types** - regular-spiking excitatory pyramidal cells and fast-spiking / low-threshold inhibitory interneurons, each with distinct intrinsic parameters.
- **Plasticity** - spike-timing-dependent plasticity (excitatory and inhibitory), short-term plasticity, homeostatic scaling, and three-factor reward/stress learning.
- **Structure** - Dale's law, an excitatory/inhibitory microcircuit, and optional columns.
- **Chemistry** - an extensible neurochemical layer (dopamine, serotonin, noradrenaline, acetylcholine, cortisol, and others) with release and clearance kinetics.
- **Instrumentation** - analyzers for oscillations, criticality, spike statistics, cell assemblies, and a battery of consciousness-correlate markers.

Two connectivity backends are provided: a dense N-by-N weight matrix (simple, exact) and a sparse edge list whose cost scales with the number of synapses rather than N squared. The sparse backend reproduces the dense one exactly (Section 8.3).

2. Neuron model

2.1 Membrane dynamics (leaky integrate-and-fire)

Each neuron holds a membrane potential V (in mV). Between spikes it integrates leak, synaptic, adaptation, external, and noise currents:

$$dV/dt = (V_{\text{rest}} - V)/\tau_m + I_{\text{syn}}/C_m - w_{\text{adapt}}/C_m + I_{\text{ext}} + \text{noise}$$

A spike is emitted when V crosses the threshold V_{thresh} ; V is then reset to V_{reset} and the neuron is held refractory for t_{ref} . Default constants follow cortical values: $V_{\text{rest}} = -70$, $V_{\text{thresh}} = -55$, $V_{\text{reset}} = -75$ mV, $\tau_m = 20$ ms, $t_{\text{ref}} = 2$ ms. Integration is explicit Euler with $dt = 0.1$ ms. Noise is scaled by dt by default; an option scales it by $\sqrt{dt}$, which is the correct scaling for a stochastic (Euler-Maruyama) update and lets modest noise drive realistic spontaneous firing (Section 8.2).

2.2 Spike-frequency adaptation

Real neurons fire quickly then slow down: a calcium- and sodium-activated potassium current hyperpolarizes the cell after each spike. This is modelled by an adaptation variable w_{adapt} (Brette and Gerstner 2005) that decays toward a subthreshold drive and is incremented on each spike:

$$dw_{\text{adapt}}/dt = (a(V - V_{\text{rest}}) - w_{\text{adapt}})/\tau_w ; \quad \text{on spike: } w_{\text{adapt}} += b$$

It is on by default because it is nearly universal in cortex and stabilizes firing.

2.3 Adaptive exponential integrate-and-fire (AdEx)

Optionally the hard threshold is replaced by a smooth, voltage-dependent spike-initiation term that reproduces a realistic upstroke and, together with the adaptation parameters, the main cortical firing patterns (tonic, adapting, bursting; Naud et al. 2008):

$$I_{\text{exp}} = (\Delta T / \tau_m) * \exp((V - V_T) / \Delta T)$$

The exponential is clamped at the peak voltage to prevent numerical overflow, and a spike is detected when V reaches the peak.

2.4 NMDA receptors

Excitatory transmission has a fast component and a slow NMDA component whose conductance is gated by voltage through the magnesium block (Jahr and Stevens 1990). Near rest the channel is closed; depolarization opens it, so NMDA acts as a coincidence detector and supplies the slow recurrent drive needed to sustain persistent activity:

$$B(V) = 1 / (1 + [\text{Mg}] * \exp(-0.062 * V) / 3.57)$$

2.5 Gap junctions and dendrites

Gap junctions are electrical connections that pass current directly between coupled neurons; wired among fast-spiking interneurons they sharpen synchrony (Galarreta and Hestrin 1999). **Dendrites:** a two-compartment option gives each neuron a separate dendritic compartment that receives excitation, integrates it, and fires regenerative dendritic spikes that then drive the soma (Poirazi and Mel 2003; Larkum 2013). This makes a single neuron a two-stage nonlinear unit (see the supralinear-summation result, Section 8.11).

2.6 Cell types

Neuron parameters can be set per neuron. Three cortical types are provided:

Type	Class	τ_m (ms)	t_{ref} (ms)	Adaptation	Role
RS	excitatory	20	2.0	strong	regular-spiking pyramidal
FS	inhibitory	10	1.0	none	fast-spiking PV+ interneuron
LTS	inhibitory	15	2.0	weak	low-threshold SST+ interneuron

3. Synapses and connectivity

3.1 Transmission

A spike from neuron i adds current to each target j , scaled by the weight $W[i,j]$ and the sign of neuron i (Dale's law: excitatory +1, inhibitory -1). Two forms are available: current-based (net drive is $g_e - g_i$) and conductance-based, which uses reversal potentials so inhibition is divisive (shunting):

$$I_{\text{syn}} = g_e * (E_{\text{exc}} - V) + g_i * (E_{\text{inh}} - V) \quad [\text{conductance-based}]$$

Conductances decay with separate excitatory and inhibitory time constants. Axonal conduction delays (finite propagation time) are optional and drawn per axon.

3.2 Backends

The dense backend stores W as an N -by- N matrix; propagation is a matrix-vector product. The sparse backend stores an edge list (pre, post, weight); propagation is a scatter-add over edges, with cost proportional to the number of synapses. The sparse backend is exact with respect to the dense one (Section 8.3) and removes the N -squared memory ceiling.

3.3 Growth and pruning

Connectivity self-organizes. A growth cone periodically computes pairwise correlations of recent spike trains and forms synapses between neurons whose activity is correlated above a threshold, modelling activity-dependent axon guidance. Weak synapses are pruned. This is the mechanism by which structure emerges from dynamics rather than being imposed.

4. Network structure (cortical microcircuit)

Beyond a homogeneous random network, a structured cortex can be built: the canonical excitatory/inhibitory microcircuit with distinct connection probabilities and strengths for the four pathways (E to E, E to I, I to E, I to I), optionally partitioned into columns with dense intra-column and sparse inter-column wiring. Fast-spiking interneurons providing fast feedback inhibition to excitatory cells generate gamma-band oscillations by the pyramidal-interneuron (PING) mechanism (Cardin et al. 2009; Buzsaki and Wang 2012). This is the first rhythm that a homogeneous network cannot produce; it requires the distinct, fast, non-adapting interneuron class wired into feedback inhibition (result in Section 8.10).

5. Plasticity and learning

5.1 Spike-timing-dependent plasticity (STDP)

The core learning rule follows Bi and Poo (1998), implemented as an online pair rule (Morrison et al. 2008). Each neuron keeps a decaying trace of its spikes. When the post-synaptic neuron fires and the pre-synaptic neuron fired shortly before (pre-before-post), the synapse is strengthened (LTP); the reverse order weakens it (LTD):

```
post fires: W[pre,post] += A_plus * trace_pre[pre]      (LTP)
pre  fires: W[pre,post] -= A_minus * trace_post[post]   (LTD)
```

Correctness note. An earlier version had this rule sign-inverted: pre-before-post depressed the synapse, making learning anti-Hebbian and contradicting the cited rule. A direct two-neuron test confirmed the inversion (a pre-then-post pair moved the weight from 0.500 to 0.489, i.e. depression where potentiation was required). The corrected rule potentiates pre-before-post and depresses post-before-pre, with traces registered after the weight change (correct pair ordering). This changes any result that depends on plasticity.

5.2 Inhibitory STDP

Inhibitory synapses follow a separate rule (Vogels et al. 2011) that grows inhibition onto over-active cells and shrinks it onto quiet ones, driving each cell toward a target rate. This establishes detailed excitatory/inhibitory balance and counteracts runaway excitation.

5.3 Short-term plasticity

Synaptic strength also changes on the 10-to-100 ms scale from vesicle depletion (depression) and residual calcium (facilitation), following Tsodyks and Markram (1997). Per axon, a utilization variable u facilitates and a resource variable x depletes on each spike; the released efficacy is $u \cdot x$. Facilitation is the mechanism for working memory held in synapses rather than in persistent firing (Mongillo et al. 2008).

5.4 Homeostatic scaling

Slow multiplicative rescaling of a neuron's incoming weights toward a target firing rate (Turrigiano and Nelson 2004) keeps STDP from collapsing the network into silence or saturation.

5.5 Three-factor reward and stress learning

For reinforcement, STDP does not edit weights directly; it lays down a slow eligibility trace that tags recently active (causal) synapses. A later neuromodulator converts the tag into a weight change (Fremaux and Gerstner 2016; Izhikevich 2007):

```
dw = lr * M * eligibility ;    M = da_gain*(dopamine - 1) - cort_gain*cortisol
```

Dopamine (reward) potentiates the tagged pathway; cortisol (a slow stress hormone, Joels and Krugers 2007) depresses it and, when sustained, impairs plasticity. The API is `reward()`, `punish()`, and `feedback(correct)`. One subtlety was found and fixed: the eligibility tag must be the causal (LTP-shaped) component, not the net STDP change, because with A_{minus} over A_{plus} the net tag is negative under synchronous activity, which would invert the sign of reward and punishment.

6. Neurochemistry

Neuromodulation is provided by an extensible chemical layer. Each chemical is an object with a baseline level, a clearance time constant, and a set of effects on five control axes that the simulator reads (dopamine, serotonin, noradrenaline, acetylcholine, cortisol). Because a chemical influences the network only by pushing these axes, the layer is a drop-in replacement for the simple five-scalar modulator, and new chemicals affect the dynamics without changing the core. The default set models the five primaries plus histamine, adenosine (sleep pressure), melatonin, oxytocin, and endorphin, with fast clearance for dopamine and slow clearance for the hormones. Verified behaviour is in Section 8.9.

7. Measurement and analysis tools

The following analyzers read activity without changing it. They exist to test whether the network is in a biologically realistic regime.

- **Oscillation analyzer** - treats the population firing rate as a local-field-potential proxy and reports power in the delta, theta, alpha, beta, and gamma bands, the dominant band, and the peak frequency (Buzsaki and Draguhn 2004).
- **Criticality analyzer** - the branching ratio (average number of descendant spikes per spike; near 1 at criticality) and neuronal-avalanche statistics (Beggs and Plenz 2003).
- **Spike statistics** - inter-spike-interval irregularity (coefficient of variation; near 1 is Poisson-like) and the Fano factor, indicating the asynchronous-irregular cortical regime (Softky and Koch 1993; Brunel 2000).
- **Cell-assembly detection** - groups of synchronously firing, mutually connected neurons read out as emergent assemblies (Hebb 1949).

A separate battery estimates operational correlates of awareness (Section 10).

8. Simulations and results

Each mechanism was checked against a known-correct expectation before being kept. Fast checks run as an automated test suite (9 of 9 passing). Longer learning runs are reproduced by demonstration functions. The tables below give the measured numbers.

8.1 Spike-timing plasticity (direction)

Test	Expectation	Measured
Pre-before-post pair	potentiate (LTP)	dW = +0.0095
Post-before-pre pair	depress (LTD)	dW = -0.0114

Confirms the corrected causal direction. The earlier inverted rule produced the opposite signs.

8.2 Single-neuron biophysics

Mechanism	Expectation	Measured
Spike-freq. adaptation	fewer spikes vs off	1318 -> 1189 spikes
AdEx	fires, numerically stable, adapts	stable; more adaptation -> fewer spikes
NMDA gate B(V)	closed at rest, open when depolarized	B(-70 mV)=0.044, B(0 mV)=0.781
sqrt(dt) noise	modest noise drives spontaneous firing	silent with dt-scaling; fires with sqrt(dt)

8.3 Sparse backend equivalence and scale

Check	Result
Propagation, dense vs sparse	match to 5e-7 (floating-point reduction order)
STDP per-edge weight update	exact, 0.0 difference
Spike trains, matched wiring	100% identical over 150 steps
Memory at N=3000 (439k edges)	7 MB (sparse) vs 72 MB (dense); grows as N ² for dense

8.4 Short-term plasticity

Regime	Expectation	Measured across a spike train
Depression	efficacy falls	0.75 -> 0.013
Facilitation	utilization u rises	0.28 -> 0.67

8.5 Network regime diagnostics

Diagnostic	Expectation	Measured
Branching ratio (constant input)	1.0 (critical)	1.000
Branching ratio (0.8 geometric decay)	0.8	0.798
ISI CV (clockwork firing)	near 0	0.00
ISI CV (Poisson firing)	near 1	0.88
Oscillation band powers	sum to about 1 in range	sum = 0.99

8.6 Reward and stress learning (three-factor)

An input group A was stimulated repeatedly and then given a teaching signal. Weight is the mean of the A-to-A synapses; identical on the dense and sparse backends.

Condition	A-to-A weight
No teaching signal	0.049 -> 0.049 (no drift; learning needs a signal)
Reward (dopamine)	0.049 -> 0.295 (potentiated)
Punishment (cortisol)	0.049 -> 0.000 (depressed)
Idle group during reward	0.048 -> 0.048 (untouched; credit is specific)

8.7 Neurochemistry

Test	Measured
Histamine raises the arousal axis	1.00 -> 1.30
Adenosine lowers arousal and attention	both fall below 0.95
Oxytocin buffers cortisol	cortisol axis reduced
Clearance kinetics	dopamine fast (2.0->1.08 in 500 ms); cortisol slow (2.0->1.87 in 200 ms)

8.8 Closed-loop learning from self-generated feedback

The network's own decoded output decides whether it was right, and reward or stress is applied automatically. Operant conditioning (rewarding the network for producing a target output on cue; Fetz 1969) learns reliably. Two-way discrimination learns once three ingredients are combined: winner-take-all competition between output groups, a dopamine reward-prediction-error baseline (Schultz et al. 1997), and a dedicated plastic input-to-output pathway (R-STDP; Izhikevich 2007; Legenstein et al. 2008).

Task	Result
Operant conditioning	response rate 0.27 -> 1.00 (learns to respond on cue)
Two-way discrimination	chance -> about 0.85 accuracy mid-run, then a late instability

The late instability is unregulated R-STDP (positive feedback and weight saturation); inhibitory STDP or homeostatic scaling are the intended stabilizers. With one of the three ingredients removed, discrimination stays at chance.

8.9 Gamma oscillations (structured cortex, PING)

Measurement	Result (N = 800, excitatory cells driven)
Dominant band	gamma (relative power 0.53); rhythm at about 24-30 Hz
Fast-spiking interneuron rate	160 Hz (characteristic PV+ signature)
Regular-spiking pyramidal rate	54 Hz

8.10 Dendritic computation (supralinear summation)

Two input groups each drove a target neuron; the somatic response was measured to each group alone and to both together.

Neuron	Group A	Group B	A + B (linear)	A and B (measured)
Active dendrite	4 spikes	4 spikes	8 expected	10 (supralinear, ratio 1.25)
Point neuron	0	0	0	0 (cannot detect the input)

The active dendrite sums supralinearly (the dendritic-spike signature); the point neuron is silent to the same distributed weak input.

9. Performance and hardware

Throughput was measured with the sparse backend at $dt = 0.1$ ms. Real time means 10,000 steps per second. Hardware: a laptop CPU and an NVIDIA RTX 3070 Laptop GPU (8.6 GB).

Device	Neurons N	Synapses	Steps/s	Fraction of real time
CPU	1,000	49k	261	2.6%
CPU	3,000	439k	29	0.29%
CPU	8,000	3.1M	5.7	0.06%
RTX 3070	3,000	439k	73	0.73%
RTX 3070	10,000	4.9M	7.5	0.08%
RTX 3070	30,000	43.9M	0.7	0.01%

The limit is speed, not memory, and the cause is the per-step Python loop and two $O(N^2)$ analyzers (the growth cone and assembly detection), which leave the GPU mostly idle. Memory (sparse backend) is about 24 bytes per synapse. At a biological 1,000 synapses per neuron this is about 24 KB per neuron:

Neurons	Synapses	Memory	Fits on
100,000	1e8	2.4 GB	RTX 3070 (8 GB)
1,000,000 (insect scale)	1e9	24 GB	RTX 4090 (24 GB) / A100
10,000,000	1e10	240 GB	multi-GPU node

On the RTX 3070, the practical range for the science (marker experiments, closed-loop learning, oscillations, criticality) is roughly 1,000 to 10,000 neurons. Reaching insect-scale networks requires both a 24 GB or larger GPU and a rewrite of the per-step core into fused GPU kernels; the current Python step loop, not the card, is the binding constraint.

10. Correlates of awareness (scope and limits)

A separate battery estimates operational markers that consciousness science uses to distinguish conscious from unconscious brain states, and validates them by requiring an awake, integrated network to separate from unconscious control conditions (anaesthetized, disconnected, structure-shuffled). This is an explicit and important limit: no measurement proves subjective experience. A high score means the network shows the dynamical signatures that, in real brains, accompany conscious states; it does not mean the network feels anything.

- **Global ignition** - a strong stimulus triggers sustained, network-wide activity while a weak one stays local (global-workspace theory; Dehaene and Changeux 2011).
- **Perturbational complexity** - the spatiotemporal complexity of the response to a brief perturbation, using Lempel-Ziv complexity (Casali et al. 2013).
- **Information integration** - lagged mutual information between network halves (a tractable proxy, not the full measure).
- **Predictive mismatch** - habituation to a repeated stimulus and a rebound to a novel one.

Validation to date: the awake network ignites (post-stimulus sustained activity 0.023) while the disconnected control does not (0.000), and complexity ordering is correct. On the question of hardware for a minimal ('bare') awareness: the honest biological reference point is insect scale, on the order of one million neurons (Barron and Klein 2016). That is a 24 GB memory target and, more importantly, requires the compute rewrite above; scale is a necessary but not sufficient condition, and none of it settles whether anything is felt.

11. Optimization roadmap

The Dragon Hatchling architecture (BDH; Kosowski et al. 2025) is a GPU-efficient, trainable model built from the same ingredients as BioNeuron (Hebbian fast weights, sparse positive activity, excitatory/inhibitory circuits). Its central efficiency trick - never form the N-by-N synaptic operator, factor it through a low-rank bottleneck - does not transfer to BioNeuron's synapses, which are structured-sparse, signed, thresholded, and NMDA-nonlinear; forcing that factorization would replace the model rather than optimize it. What does transfer is narrower and lives in the analysis and execution layers, not the biophysics:

- **Active-set assembly detection** - the co-firing matrix is a correlation of spike rows, and a silent neuron contributes exactly zero, so restricting the computation to neurons that fired is bit-exact and removes the dominant cost past a few thousand neurons.
- **Sync-free stepping** - deferring the per-step device-to-host reads (about 5 per step) into once-per-chunk reads is bit-identical and gives roughly 1.5 to 3x at moderate N.
- **Optional low-rank sketch** of the correlation matrices removes the N-by-N allocation that limits the 8 GB GPU; it is approximate, so it is opt-in.

Not advisable: making the synapses themselves globally low-rank (it changes the dynamics), or a factored eligibility trace on the dense backend (it is more compute, not less). A differentiable, backprop-trainable readout is possible but is a separate machine-learning capability, not a fidelity feature, and would be an opt-in module.

12. Software and reproducibility

The project is a pip-installable package, **bioneuron** (v1.8, MIT). Modules: core (neurons, synapses, plasticity, growth, analyzers, orchestration); sparse_synapses (edge-list backend); neurochemistry (chemical layer); cortex (cell types and microcircuit); closed_loop (self-feedback tasks). A fast test suite (9 of 9 passing) covers the spike-timing direction, sparse/dense equivalence, adaptation, inhibitory STDP, neurochemistry axes, reward and stress learning, cell-type firing, and dendritic supralinearity. Every result in Section 8 is reproduced by these tests or by the demonstration functions.

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